VFS DBVF 26-27 - Mission Model Results

Full-success mission (FM-1 + FM-2 x3 payloads + FM-3 x3 retrievals, end at Home) on one 6S pack

Airframe	Quadcopter, 10 lb unloaded incl. motors, props, battery (R = 1.4); 3 lb water payload
Drag / speed	Cd 1.0 (quadcopter frontal area), 15 m/s governor
Mission	13 takeoffs, 13 landings, 16 transits, 3 hover drops; all transits at 30 ft AGL
Time limit	600 s; throttle chosen per phase for minimum energy inside it
Voltage	6S - every motor curve measured at ~24 V, no pack stacking
Sweep	63 motor/prop combos x 9 batteries = 533 runs
'Usable' means	flies the whole mission, loaded thrust/weight >= 1.8, pack capacity x C covers 4-motor peak current

Recommendation

- Battery: Tattu RLine5 2200 6S 150C (346 g). 1300 mAh finishes too, but with no real reserve.
- Most efficient: Vertiq 60-08 300KV P22x6.6 - 70% of the pack, thrust/weight 3.4, 303 s.
- Best US-brand, published price: KDE4014XF-380 18.5in dual - 85% of the pack, $620 for 4 motors.
- Every usable combo finishes in ~298-308 s: ~292 s of margin on the 600 s limit.

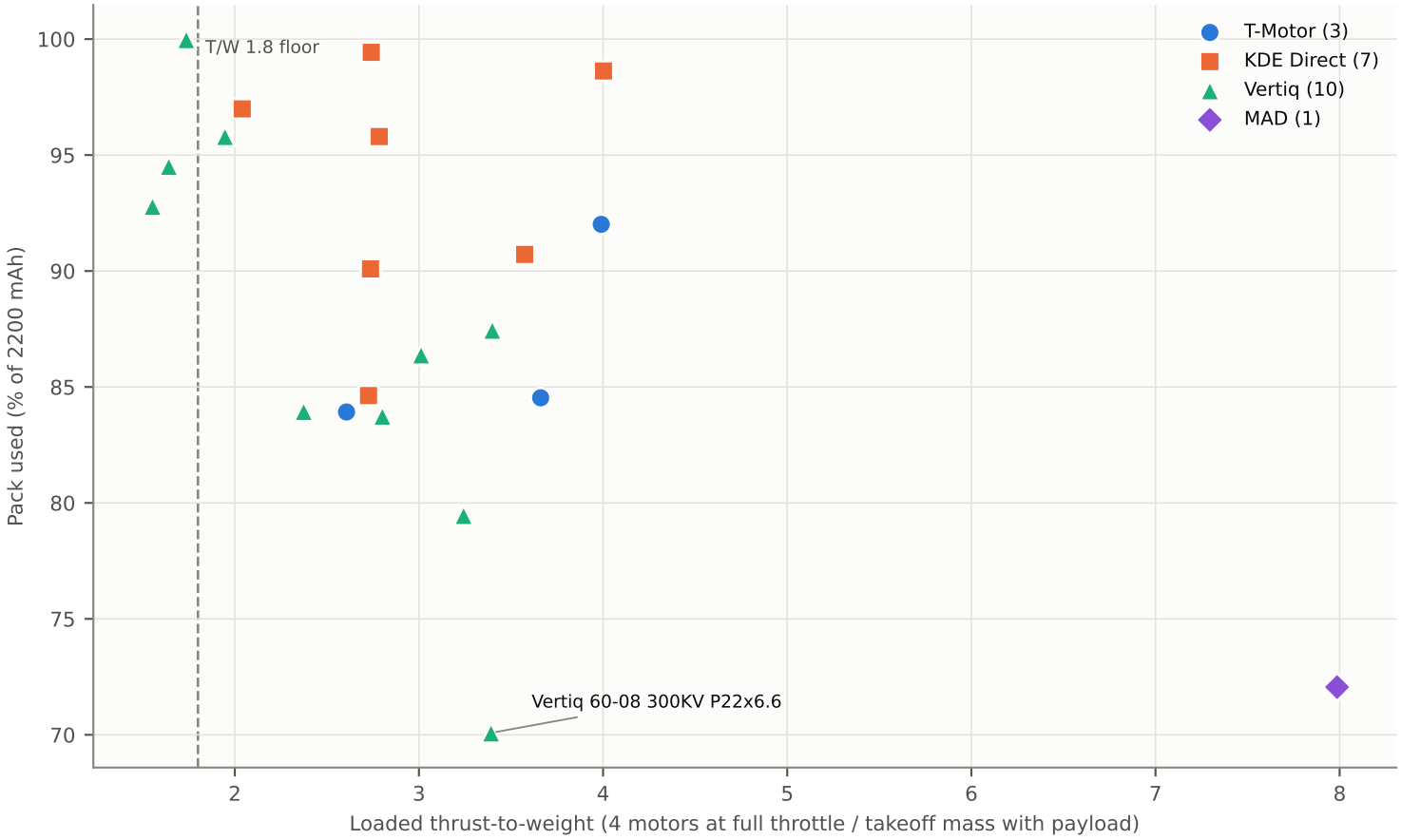
Top 15 motor/prop combos on the RLine5 2200 mAh pack

Motor + prop	Prop in	Mass kg*	T/W loaded	Time s	Energy mAh	Pack used	Peak A	4 motors
Vertiq 60-08 300KV P22x6.6	22	4.54	3.39	303	1542	70%	133	quote
MAD V62 280KV 12S Fluxer 18.1x7.2	18.1	4.54	7.99	299	1585	72%	321	$512
Vertiq 60-08 300KV P20x6	20	4.54	3.24	304	1748	79%	140	quote
Vertiq 60-08 300KV P18x6.1	18	4.54	2.80	304	1842	84%	114	quote
T-Motor MN6007II P18x6.1 (est.)	18	4.54	2.61	302	1846	84%	100	$727
Vertiq 60-08 300KV P17x5.8	17	4.54	2.37	304	1847	84%	90	quote
T-Motor MN6007II P21x6.3	21	4.54	3.66	303	1860	85%	155	$727
KDE4014XF-380 18.5in dual	18.5	4.54	2.73	307	1862	85%	124	$620
Vertiq 60-08 660KV P20x6	20	4.54	3.01	304	1900	86%	131	quote
Vertiq 60-08 660KV P18x6.1	18	4.54	3.40	305	1924	87%	165	quote
KDE4213XF-360 18.5in dual	18.5	4.54	2.74	306	1982	90%	125	$680
KDE5215XF-330 18.5in dual	18.5	4.54	3.57	304	1996	91%	164	$1040
T-Motor MN6007II P22x6.6	22	4.54	3.99	298	2024	92%	177	$727
KDE5215XF-330 15.5in triple	15.5	4.54	2.78	307	2108	96%	132	$1040
Vertiq 60-08 300KV P16x5.4	16	4.54	1.95	304	2108	96%	76	quote

*takeoff mass without payload. Vertiq mass includes its built-in ESC; others exclude the ESC.

Trade space: efficiency vs. thrust margin

Every combo that flies the mission on the 2200 mAh pack. Lower = less battery; right = more thrust headroom.



Best usable combo at each prop size

Prop in	Motor + prop	T/W loaded	Pack used	Peak A
15.5	KDE5215XF-330 15.5in triple	2.78	96%	132
16	Vertiq 60-08 300KV P16x5.4	1.95	96%	76
17	Vertiq 60-08 300KV P17x5.8	2.37	84%	90
18	Vertiq 60-08 300KV P18x6.1	2.80	84%	114
18.1	MAD V62 280KV 12S Fluxer 18.1x7.2	7.99	72%	321
18.5	KDE4014XF-380 18.5in dual	2.73	85%	124
20	Vertiq 60-08 300KV P20x6	3.24	79%	140
21	T-Motor MN6007II P21x6.3	3.66	85%	155
22	Vertiq 60-08 300KV P22x6.6	3.39	70%	133

Battery sizing for the top picks

Energy left after the full mission. 'no' = can't finish, loaded T/W < 1.8, or C rating too low.

Pack	Pack g	Vertiq 60-08 300KV P22x6.6	T-Motor MN6007II P18x6.1K	KDE4014XF-380 18.5x5.5	SunnyV62 280KV 12S Fluxer 18.1x7.2
Tattu FunFly 1300 100C	225	no	no	no	no
Tattu RLine5 2200 150C	346	658 (30%)	354 (16%)	338 (15%)	615 (28%)
Tattu 4500 25C	670	no	2654 (59%)	no	no
GensAce GTech 5500 60C	828	3958 (72%)	3654 (66%)	3638 (66%)	3915 (71%)
Tattu GTech 8000 25C	1160	6458 (81%)	6154 (77%)	6138 (77%)	no
Tattu Plus 10000 25C	1517	8458 (85%)	8154 (82%)	8138 (81%)	no
Tattu Plus 12000 15C	1670	10458 (87%)	10154 (85%)	10138 (84%)	> 10 lb
Tattu GTech 22000 30C	2460	20458 (93%)	20154 (92%)	20138 (92%)	> 10 lb
Tattu GTech 30000 25C	3673	> 10 lb	28154 (94%)	28138 (94%)	> 10 lb

Reading it: land with >= 20-30% left in practice - the model drains to 0% and does not model voltage sag.

Aircraft is held at 10 lb unloaded: a bigger pack doesn't cut T/W, it eats frame budget.

'> 10 lb' = motors + pack alone exceed 10 lb.

Course as modeled

Aim points from the 26-27 field drawing, feet from Home (H). Pickup spot, A/B/C inset and D center scaled from the picture.

Point	x ft	y ft
H	0	0
S	0	-20
L	-100	0
A	-285	10
B	-285	0
C	-285	-10
D	-317	0

#	Leg	
1	takeoff	
2	cruise	317 ft
3	land	
4	takeoff	
5	cruise	317 ft
6	land	
7	takeoff	
8	cruise	20 ft
9	hover	15 s
10	land	
11	add_payload	
12	takeoff	
13	cruise	287 ft
14	hover	10 s
15	remove_payload	
16	cruise	287 ft
17	hover	15 s
18	land	
19	add_payload	
20	takeoff	
21	cruise	286 ft
22	hover	10 s
23	remove_payload	
24	cruise	286 ft
25	hover	15 s
26	land	
27	add_payload	
28	takeoff	
29	cruise	285 ft
30	hover	10 s
31	remove_payload	
32	cruise	285 ft

#	Leg	
33	land	
34	takeoff	
35	cruise	285 ft
36	hover	15 s
37	land	
38	add_payload	
39	takeoff	
40	cruise	185 ft
41	land	
42	remove_payload	
43	takeoff	
44	cruise	185 ft
45	hover	15 s
46	land	
47	add_payload	
48	takeoff	
49	cruise	185 ft
50	land	
51	remove_payload	
52	takeoff	
53	cruise	185 ft
54	hover	15 s
55	land	
56	add_payload	
57	takeoff	
58	cruise	185 ft
59	land	
60	remove_payload	
61	takeoff	
62	cruise	100 ft
63	land	

All combos on the 2200 mAh pack (1)

Sorted by energy used. Full per-battery results: vfs_dbvf_26-27_results.csv

Motor + prop	Prop in	T/W loaded	Time s	Pack used	Peak A	Usable
Vertiq 60-08 300KV P22x6.6	22	3.39	303	70%	133	yes
MAD V62 280KV 12S Fluxer 18.1x7.2	18.1	7.99	299	72%	321	yes
Vertiq 60-08 300KV P20x6	20	3.24	304	79%	140	yes
Vertiq 60-08 300KV P18x6.1	18	2.80	304	84%	114	yes
T-Motor MN6007II P18x6.1 (est.)	18	2.61	302	84%	100	yes
Vertiq 60-08 300KV P17x5.8	17	2.37	304	84%	90	yes
T-Motor MN6007II P21x6.3	21	3.66	303	85%	155	yes
KDE4014XF-380 18.5in dual	18.5	2.73	307	85%	124	yes
Vertiq 60-08 660KV P20x6	20	3.01	304	86%	131	yes
Vertiq 60-08 660KV P18x6.1	18	3.40	305	87%	165	yes
KDE4213XF-360 18.5in dual	18.5	2.74	306	90%	125	yes
KDE5215XF-330 18.5in dual	18.5	3.57	304	91%	164	yes
T-Motor MN6007II P22x6.6	22	3.99	298	92%	177	yes
Vertiq 40-14 400KV P18x6.1	18	1.55	306	93%	57	fails T/W
Vertiq 40-14 400KV P17x5.8	17	1.64	306	95%	64	fails T/W
KDE5215XF-330 15.5in triple	15.5	2.78	307	96%	132	yes
Vertiq 60-08 300KV P16x5.4	16	1.95	304	96%	76	yes
KDE4213XF-360 15.5in dual	15.5	2.04	306	97%	84	yes
KDE5215XF-330 18.5in triple	18.5	4.00	304	99%	211	yes
KDE3520XF-400 18.5in dual	18.5	2.74	308	99%	158	yes
Vertiq 40-14 400KV P16x5.4	16	1.74	307	100%	73	fails T/W
KDE4213XF-360 18.5in triple	18.5	2.93	-	-	148	can't fly
KDE3520XF-400 15.5in dual	15.5	2.30	-	-	110	can't fly
KDE4014XF-380 18.5in triple	18.5	2.93	-	-	148	can't fly
Vertiq 60-08 660KV P17x5.8	17	3.55	-	-	191	can't fly
KDE4215XF-465 18.5in dual	18.5	3.80	-	-	250	can't fly
KDE4014XF-380 15.5in dual	15.5	1.98	-	-	84	can't fly
Vertiq 60-08 660KV P16x5.4	16	3.47	-	-	197	can't fly
Vertiq 40-14 400KV P15x5	15	1.74	-	-	76	can't fly
KDE4213XF-360 15.5in triple	15.5	2.23	-	-	102	can't fly
KDE4014XF-380 15.5in triple	15.5	2.20	-	-	103	can't fly
KDE5215XF-330 15.5in dual	15.5	2.34	-	-	102	can't fly
KDE3520XF-400 15.5in triple	15.5	2.42	-	-	135	can't fly
KDE4215XF-465 15.5in dual	15.5	3.18	-	-	180	can't fly
KDE4215XF-465 15.5in triple	15.5	3.30	-	-	214	can't fly
KDE5215XF-435 18.5in dual	18.5	4.94	-	-	307	can't fly
Vertiq 60-08 660KV P15x5	15	3.91	-	-	248	can't fly
Vertiq 40-06 370KV P15x5	15	1.09	-	-	40	can't fly
Vertiq 40-14 400KV P14x4.8	14	1.53	-	-	69	can't fly
Vertiq 40-06 370KV P16x5.4	16	1.07	-	-	38	can't fly

All combos on the 2200 mAh pack (2)

Sorted by energy used. Full per-battery results: vfs_dbvf_26-27_results.csv

Motor + prop	Prop in	T/W loaded	Time s	Pack used	Peak A	Usable
Vertiq 40-06 370KV P14x4.8	14	1.15	-	-	46	can't fly
Vertiq 60-08 660KV P14x4.8	14	4.40	-	-	308	can't fly
Vertiq 40-06 770KV P14x4.8	14	1.15	-	-	48	can't fly
KDE5215XF-435 15.5in dual	15.5	3.55	-	-	205	can't fly
KDE3520XF-400 12.5in triple	12.5	1.26	-	-	54	can't fly
Vertiq 40-06 770KV P13x4.4	13	1.43	-	-	73	can't fly
KDE4014XF-380 12.5in triple	12.5	1.11	-	-	44	can't fly
KDE5215XF-435 15.5in triple	15.5	4.02	-	-	251	can't fly
Vertiq 40-14 400KV P13x4.4	13	1.12	-	-	51	can't fly
KDE3520XF-400 12.5in dual	12.5	1.10	-	-	45	can't fly
KDE4215XF-465 12.5in dual	12.5	1.53	-	-	72	can't fly
Vertiq 40-06 770KV P12x4	12	1.31	-	-	66	can't fly
Vertiq 60-08 660KV P13x4.4	13	3.49	-	-	251	can't fly
KDE4215XF-465 12.5in triple	12.5	1.67	-	-	85	can't fly
KDE2814XF-515 12.5in dual	12.5	1.58	-	-	88	can't fly
KDE2814XF-515 12.5in triple	12.5	1.69	-	-	100	can't fly
Vertiq 60-08 660KV P12x4	12	3.10	-	-	222	can't fly
KDE4213XF-360 12.5in triple	12.5	1.02	-	-	41	can't fly
Vertiq 40-06 370KV P17x5.8	17	0.99	-	-	33	can't fly
Vertiq 40-06 370KV P13x4.4	13	1.00	-	-	37	can't fly
Vertiq 40-06 370KV P12x4	12	0.89	-	-	34	can't fly
KDE4213XF-360 12.5in dual	12.5	0.90	-	-	33	can't fly
KDE4014XF-380 12.5in dual	12.5	0.93	-	-	37	can't fly

Data sources & caveats

Motor curves (all measured at ~24 V / 6S):

T-Motor MN6007II KV320 - T-Motor bench sheet, P21x6.3 and P22x6.6.

T-Motor MN6007II + P18x6.1 - ESTIMATED by estimate_prop_curve.py from the 21/22 in data (method checked on KDE data: thrust within 6% at 50-100% throttle; current padded +11%).

KDE Direct - KDE performance sheets (tested at 3,500 ft; sea-level thrust ~10% higher).

Vertiq 40-06 / 40-14 / 60-08 - vertiq.co/thrust-data, 24 V, continuous-rated rows only.

Batteries - manufacturer net weight and genstattu.com list price (Sept 2026).

Props - KDE blade weights; T-Motor P-series weights/prices from ligpower.com.

Caveats:

Energy is drained to 0%; real packs sag under load - keep a 20-30% reserve.

Throttle is never modeled below each sheet's lowest point (40% for T-Motor), so light-load hover current is over- rather than under-estimated.

Landing is a prescribed descent profile, not optimized.

Coaxial losses, wind, and control margin beyond thrust/weight are not modeled.

Vertiq prices are quote-only (shown as 'quote').

Reproduce:

```
python VFS_DBVF_26-27.py --data-dir Motors --battery-file Batteries \
    --vehicle-file vechicle_configs --no-plot --grid-step 1
python make_result_sheet.py
```